REMOTE DEPOSIT CAPTURE — MOBILE APP STORY MAP

Includes all section 18 functions, explicit ERROR CONDITIONS, FINGERPRINT AUTHENTICATION and TOUCH ID AUTHENTICATION.

LEFT TO RIGHT = chronological activities. TOP TO BOTTOM = steps, error/recovery variations, then optional enhancements.

Phone platforms and sign-in methods are alternatives: each customer chooses the supported route rather than completing every option.

	1. Install and prepare	2. Sign in securely	3. View accounts and choose	4. Capture the check	5. Enter and validate	6. Review and submit	7. Track deposit and get help	8. Explore offers
MAJOR ACTIVITIES								
MAIN STEPS Core functionality Read steps in each column from top to bottom.	S01 Download for iPhone	S05 Use portal credentials	S08 View accounts and balances	S10 Prepare camera and check	S14 Enter check amount	S17 Review account, amount, images	S19 Receive acknowledgement	S21 View bank offers
	S02 Download for Android	S06 Use an approved alternative	S09 Select eligible deposit account	S11 Take check-front picture	S15 Enter required check details	S18 Confirm and send e-check	S20 Refresh status and balance	S22 Open promotion details
	S03 Download for Windows Phone	S07 Sign out; lock idle sessions		S12 Take check-back picture	S16 Match images against details			
	S04 Check remote-capture readiness			S13 Preview and retake securely				
ERROR / RECOVERY Core functionality Return to the same step after correction.	E01 Setup or eligibility problem	E02 LOGIN FAILURE: retry or recover	E04 Balance unavailable: show stale state	E06 Camera denied: enable or cancel	E08 Invalid amount or missing fields	E10 Network timeout: check outcome	E12 Rejected/pending: show next action	E13 Offers unavailable: continue banking
		E03 Expired code: request a new one	E05 Ineligible account: choose another	E07 IMAGE SCANNING ERROR: retake	E09 Image/detail mismatch: correct	E11 Duplicate/fraud alert: block and explain		
ADDITIONAL OPTIONS Authentication and convenience variations included in this story map.	—	O01 FINGERPRINT authentication	O03 Remember preferred account	O04 Auto-capture with edge guides	O05 Suggest amount; always verify	O06 Resume an encrypted unsent draft	O07 View deposit history	O09 Choose personalized offers
		O02 TOUCH ID authentication					O08 Opt into status notifications	

S = normal story; E = error/recovery variation; O = optional improvement. Every card has an expanded user story and source trace in this document.

Both front AND back images, amount entry, account selection, image/detail validation, secure submission and promotions are mandatory Core scope.

Offers are an independent branch, never a deposit prerequisite. Submission acknowledgement is not a promise that funds are cleared or immediately available.

Error conditions and authentication variations

These are explicit branches of the story map. Each branch identifies the stimulus, system response and customer recovery; it is part of the submitted product scope.

Condition	Trigger	App response	Recovery	Map ID
Login failure	Wrong username/password	Deny account access; show safe login error.	Retry credentials or use the bank recovery route.	E02
Alternative authentication fails	Expired or incorrect code	Reject the code; keep the session unauthenticated.	Request a new bank-verified code.	E03
Fingerprint authentication fails	Unrecognized fingerprint, unavailable sensor or cancelled prompt	Do not log in; do not reveal account information.	Retry within bank policy or use portal credentials/verified code.	O01
Touch ID authentication fails	Touch ID mismatch, lockout, unavailable sensor or cancellation	Keep the app locked; display the permitted fallback.	Use bank-approved credentials/code; no biometric bypass.	O02
Camera or scanning error	Permission denied, blurred front/back, missing side or glare	Block image validation/submission and identify the affected side.	Enable camera or retake the required image; preserve other valid input.	E06-E07
Invalid entered details	Empty/non-numeric/non-positive amount or required field missing	Show a field-specific error and do not submit.	Correct the amount/details and validate again.	E08
Image/details mismatch	Image validation disagrees with the entered amount/details	Block deposit and explain the mismatch.	Correct entered data or retake the image, then revalidate.	E09
Network or submission error	Connection lost before a receipt is received	Show pending/unknown outcome, not confirmed success.	Look up the existing request reference before any retry.	E10
Duplicate or fraud warning	Bank flags duplicate/tampered/ineligible check	Do not show a successful deposit; present the bank result.	Review the issue or contact bank support.	E11
Account, status or offer error	Account ineligible, bank rejects check, balance/offer unavailable	Explain the problem; preserve known status and avoid false balance claims.	Choose another eligible account, refresh, contact support or skip offers.	E04-E05, E12-E13

Biometric success paths

Fingerprint: after bank-verified sign-in and explicit opt-in, a supported phone accepts a recognized fingerprint and opens the authorized session. Touch ID: after the same enrollment/consent, a supported iPhone accepts Touch ID and opens the authorized session. Both methods retain a non-biometric fallback and follow bank session/lockout rules; the bank does not receive a raw fingerprint image.

Scope check

Required functions: downloads for iPhone, Android and Windows Phone; portal or alternative sign-in; balances; front AND back capture; amount/details; destination account; image-versus-details validation; secure e-check deposit; bank offers. Fingerprint and Touch ID are additional authentication options, not replacements for basic bank verification.

Readable story map — activities, steps and variations

This is the same story map as the landscape first page, expanded as selectable text. The backbone is Install → Sign in → View/select accounts → Capture → Enter/validate → Review/submit → Track/help; offers are a separate optional customer activity. The Core includes the main steps AND error recovery; additional options include biometric authentication and convenience features.

1. Install and prepare

Customer outcome: Use the service on my phone.

Main steps — Core: S01 Download for iPhone; S02 Download for Android; S03 Download for Windows Phone; S04 Check remote-capture readiness

Error/recovery variations — Core: E01 Setup or eligibility problem

Optional enhancements — Additional option: No additional story in this release.

2. Sign in securely

Customer outcome: Access my bank identity.

Main steps — Core: S05 Use portal credentials; S06 Use an approved alternative; S07 Sign out; lock idle sessions

Error/recovery variations — Core: E02 LOGIN FAILURE: retry or recover; E03 Expired code: request a new one

Optional enhancements — Additional option: O01 FINGERPRINT authentication; O02 TOUCH ID authentication

3. View accounts and choose

Customer outcome: Know my balance and destination.

Main steps — Core: S08 View accounts and balances; S09 Select eligible deposit account

Error/recovery variations — Core: E04 Balance unavailable: show stale state; E05 Ineligible account: choose another

Optional enhancements — Additional option: O03 Remember preferred account

4. Capture the check

Customer outcome: Provide readable front/back images.

Main steps — Core: S10 Prepare camera and check; S11 Take check-front picture; S12 Take check-back picture; S13 Preview and retake securely

Error/recovery variations — Core: E06 Camera denied: enable or cancel; E07 IMAGE SCANNING ERROR: retake

Optional enhancements — Additional option: O04 Auto-capture with edge guides

Readable story map — continued

5. Enter and validate

Customer outcome: Provide correct check information.

Main steps — Core: S14 Enter check amount; S15 Enter required check details; S16 Match images against details

Error/recovery variations — Core: E08 Invalid amount or missing fields; E09 Image/detail mismatch: correct

Optional enhancements — Additional option: O05 Suggest amount; always verify

6. Review and submit

Customer outcome: Authorize one intended deposit.

Main steps — Core: S17 Review account, amount, images; S18 Confirm and send e-check

Error/recovery variations — Core: E10 Network timeout: check outcome; E11 Duplicate/fraud alert: block and explain

Optional enhancements — Additional option: O06 Resume an encrypted unsent draft

7. Track deposit and get help

Customer outcome: Understand the bank outcome.

Main steps — Core: S19 Receive acknowledgement; S20 Refresh status and balance

Error/recovery variations — Core: E12 Rejected/pending: show next action

Optional enhancements — Additional option: O07 View deposit history; O08 Opt into status notifications

8. Explore offers

Customer outcome: Discover optional bank offers.

Main steps — Core: S21 View bank offers; S22 Open promotion details

Error/recovery variations — Core: E13 Offers unavailable: continue banking

Optional enhancements — Additional option: O09 Choose personalized offers

Coverage of every section 18 mobile requirement

Requirement	Mobile capability	Map stories
18(a)	iPhone, Android, Windows Phone downloads	S01-S03
18(b)	Internet-portal credentials or other authentication	S05-S07; E02-E03; O01 fingerprint; O02 Touch ID
18(c)	View account balances	S08, S20; E04
18(d)(i)	Secure front and back check photographs	S10-S13; E06-E07; E11
18(d)(ii)	Enter amount and check details	S14-S15; E08
18(d)(iii)	Select the destination deposit account	S09, S17; E05
18(d)(iv)	Validate images against entered details	S16; E09
18(d)(v)	Deposit the e-check into the account	S17-S20; E10-E12
18(e)	Bank shares offers and promotions	S21-S22; E13; O09

Acceptance examples

• A valid deposit: sign in, view balances, choose an eligible account, capture front/back, enter amount, pass image/detail checks, review and submit once; receive a bank reference and status.

• A failed login: show a safe error, grant no account access, and offer retry or bank recovery. Expired alternative codes are rejected; biometric failures fall back to an approved method.

• A camera/image error: a missing permission, blurred image or absent check side blocks the affected step and gives a clear retry/retake route.

• A data mismatch: invalid amount or image-versus-entered-detail mismatch blocks submission until the customer corrects it.

• A submission timeout: check the existing request status before resending; an unknown result is not displayed as a successful cleared deposit.

• Banking without promotions: an unavailable offer never prevents viewing a balance or submitting a check.

Explicit assumptions and scope

The bank provides existing identity, remote-capture enrollment/eligibility, balances, check-validation and deposit-processing/status services. Alternative authentication may use a bank-approved verified code; fingerprint and Touch ID are optional, supported-device conveniences following bank authentication, with a non-biometric fallback.

Check images use protected temporary storage/transport and bank-side duplicate/tamper checks; “fraud-proof” in the vision is treated as a fraud-resistance requirement rather than an absolute guarantee. Bank acknowledgement, clearing and funds availability are distinct; the app displays bank-provided statuses and balances.

All three platforms in the fictional case remain explicit, including Windows Phone; real-world platform/store feasibility must be agreed before implementation. Desktop scanners, browser plug-ins, scanner purchases, portal account administration and bank-side clearing are outside this mobile map. Help links may use the existing bank support service; building that service is not mobile scope.

Source: supplied Case Study: Remote Deposit Capture, especially section 18(a-e), and the SampleStoryMap example. Optional improvements and detailed error behavior are reasonable product assumptions allowed by the assignment, not claims of implemented software.

Expanded customer stories

These stories provide the user need and benefit for every card on the map. Normal steps and error handling are Core scope; additional options remain explicitly part of this story map.

Story / priority	Customer need and benefit
S01 / Core	As a customer, I want to install the iPhone app so that I can deposit from my iPhone. Trace: 18(a)
S02 / Core	As a customer, I want to install the Android app so that I can deposit from my Android phone. Trace: 18(a)
S03 / Core	As a customer, I want to install the Windows Phone app specified by the case so that the third required platform is included. Trace: 18(a)
S04 / Core	As a customer, I want to check whether my existing bank account is eligible for remote capture so that I know how to prepare before depositing. Trace: 3-4; assumption
S05 / Core	As a customer, I want to sign in using my internet portal account so that I can access my existing bank identity. Trace: 18(b)
S06 / Core	As a customer, I want to use a bank-approved alternative such as a verified one-time code so that I have another supported authentication route. Trace: 18(b); assumption
S07 / Core	As a customer, I want to sign out and have idle access locked so that my account is protected on a shared or unattended phone. Trace: 5,15; assumption
S08 / Core	As a customer, I want to view bank-provided account balances with a refresh indicator so that I can understand my current funds. Trace: 18(c)
S09 / Core	As a customer, I want to choose an eligible destination account so that the deposit goes to the account I intend. Trace: 18(d)(iii)
S10 / Core	As a customer, I want to see camera and endorsement guidance so that I can prepare the check correctly. Trace: 18(d)(i); assumption
S11 / Core	As a customer, I want to capture a clear check-front image so that the bank can read the front details. Trace: 18(d)(i)
S12 / Core	As a customer, I want to capture a clear check-back image so that both required check sides are submitted. Trace: 18(d)(i)
S13 / Core	As a customer, I want to preview, retake and protect temporary check images so that unreadable or exposed photos are avoided. Trace: 18(d)(i)
S14 / Core	As a customer, I want to enter the check amount so that the bank receives my intended deposit value. Trace: 18(d)(ii)
S15 / Core	As a customer, I want to complete other bank-required check fields so that my deposit information is complete. Trace: 18(d)(ii); assumption
S16 / Core	As a customer, I want to validate captured images against the entered amount and details so that inconsistent check data is corrected before submission. Trace: 18(d)(iv)
S17 / Core	As a customer, I want to review the complete deposit summary so that I can correct mistakes before authorizing it. Trace: 18(d)(ii-v)
S18 / Core	As a customer, I want to authorize a secure submission to my selected bank account so that the bank can receive and process my e-check. Trace: 18(d)(v)
S19 / Core	As a customer, I want to receive a bank reference and receipt so that I have evidence that my request was received. Trace: 18(d)(v); assumption

Story / priority	Customer need and benefit
S20 / Core	As a customer, I want to view the bank-reported outcome and updated balance so that I can distinguish pending processing from funds available. Trace: 18(c,d)(v); assumption
S21 / Core	As a customer, I want to view current bank offers in the app so that I can discover new banking opportunities. Trace: 18(e)
S22 / Core	As a customer, I want to read offer conditions and the next action so that I can decide whether an offer is useful. Trace: 18(e)
E01 / Core	As a customer, I want to see supported-device/setup guidance when installation or eligibility fails so that I can resolve the issue or contact the bank. Trace: assumption
E02 / Core	As a customer, I want to see a safe login error and a bank recovery route after incorrect credentials so that I can recover without entering an unauthorized account. Trace: 18(b); error variation
E03 / Core	As a customer, I want to retry a failed or expired alternative identity check so that an expired code does not grant access. Trace: 18(b); error variation
E04 / Core	As a customer, I want to see when balances cannot be refreshed so that I do not mistake stale information for current funds. Trace: 18(c); error variation
E05 / Core	As a customer, I want to see why an account cannot receive the check and select another eligible account so that I can correct the destination before sending. Trace: 18(d)(iii); error variation
E06 / Core	As a customer, I want to get permission guidance when the camera is unavailable so that I can enable it or cancel safely. Trace: 18(d)(i); error variation
E07 / Core	As a customer, I want to be told which check image is missing or unreadable so that I can retake that side before continuing. Trace: 18(d)(i); error variation
E08 / Core	As a customer, I want to correct empty, nonnumeric or invalid deposit details so that invalid values are not sent as a valid deposit. Trace: 18(d)(ii); error variation
E09 / Core	As a customer, I want to correct details or retake an image when validation detects a mismatch so that the bank receives consistent information. Trace: 18(d)(iv); error variation
E10 / Core	As a customer, I want to look up the same request before retrying an interrupted submission so that I do not accidentally submit a duplicate check. Trace: 18(d)(v); error variation
E11 / Core	As a customer, I want to receive the bank validation result for suspected duplicates or tampering so that a risky check is not presented as a successful deposit. Trace: 18(d)(i,iv,v); error variation
E12 / Core	As a customer, I want to see the bank reason, pending status or rejection and reach support so that I understand the result without assuming that funds have cleared. Trace: 18(d)(v),7; error variation
E13 / Core	As a customer, I want to skip unavailable promotions so that checking balances or depositing is never blocked by offers. Trace: 18(e); error variation
O01 / Additional option	As a customer, I want to opt into supported fingerprint authentication after bank verification so that I can re-enter conveniently and fall back to a bank-approved method. Trace: rubric enhancement; assumption
O02 / Additional option	As a customer, I want to opt into Touch ID on a supported iPhone after bank verification so that I can use biometric convenience without losing a recovery route. Trace: rubric enhancement; assumption

Story / priority	Customer need and benefit
O03 / Additional option	As a customer, I want to remember a preferred account while confirming it each time so that repeat deposits take fewer steps. Trace: optional assumption
O04 / Additional option	As a customer, I want to use framing guides and optional automatic image capture so that I can take clear pictures more easily. Trace: optional assumption
O05 / Additional option	As a customer, I want to review a suggested amount before accepting it so that typing is reduced without removing image/detail validation. Trace: optional assumption
O06 / Additional option	As a customer, I want to resume a protected unsent draft after signing in again so that I can continue interrupted preparation without duplicating a submitted check. Trace: optional assumption
O07 / Additional option	As a customer, I want to review earlier deposit references and outcomes so that I can find a previous deposit without resubmitting it. Trace: optional assumption
O08 / Additional option	As a customer, I want to receive optional deposit-status notifications so that I do not have to repeatedly refresh the app. Trace: optional assumption
O09 / Additional option	As a customer, I want to opt into personalization and dismiss unwanted offers so that promotions remain relevant and under my control. Trace: optional assumption